

Predicate

Daawa karna
Tasdeek karna

What?

Property of Subject
Expression that contain variables

Why do we study it in DM? → to deal with variables

Propositional Function

Predicate is also called
Propositional Function

$P(x)$

Denote: $P(\text{variables})$

What is the truth value of $P(9)$ and $P(2)$?

If a value is assigned to the variable x ,
then $P(x)$ becomes a proposition.

He goes to school

First Part
Subject

Second Part

Predicate

Property of Subject

$x > 5$

It is not
Proposition

✓ $9 > 5$

$P(9)$ True

$P(2)$ False

✗ $2 > 5$

$7 > 5$

Proposition → Statement ; True or False

Consider the propositional function $Q(x, y, z)$ defined by:

$$x + y = z$$

What is the truth value of $Q(2, 3, 5)$?

True

What is the truth value of $Q(0, 1, 2)$?

False



$$2+3=5$$

$$5=5$$



$$0+1=2$$

$$1=2$$

تجویز، قول

Proposition

Can be true or false

Ali ki desk py book pari hai



Predicate becomes Proposition
If value is assigned to variable

Propositional function $P(x)$

$P(x) : x > 3$

$5 > 3$

$x > 3$

Ali ki desk py book pari hogi shyd

دعوی کرنا Predicate

Can not be true or false
Until value is not assigned
to variable



If $P(x)$ is $x > 3$, then what
Is truth value of $P(4)$

$4 > 3$

TRUE

QUANTIFIER

مقدار ظاہر کرنے والا

A quantifier is an operator used to create a proposition from a propositional function. An other way, if we do not assign values to $p(x)$

$$P(x) \quad x < 5$$

Values not exceeding 4

UNIVERSAL QUANTIFIER

$P(x)$ is true For all values of x in the universe of discourse

Universe of discourse
(i.e., range)

FOR ALL

$$\forall x P(x)$$

EXISTENTIAL QUANTIFICATION

There exists a value of x in the universe of discourse such that $P(x)$ is true

$$\exists x P(x)$$

What is the truth value of $\forall x P(x)$, where $P(x)$ is the statement " $x^2 < 10$ " and the universe of discourse consists of positive integers not exceeding 4?

False

$0^2 < 10$	$1^2 < 10$	$2^2 < 10$	$3^2 < 10$	$4^2 < 10$
$0 < 10$	$1 < 10$	$4 < 10$	$9 < 10$	$16 < 10$
✓	✓	✓	✓	

What is the truth value of $\exists x P(x)$, where $P(x)$ is the statement " $x^2 < 10$ " and the universe of discourse consists of positive integers not exceeding 4?

True

$$\begin{array}{l} 0^2 < 10 \\ 0 < 10 \\ \checkmark \end{array}$$

Defintion: Two quantifiers are said to be nested if one is within the scope of the other.

For example: $\forall x \exists y Q(x, y)$



$\exists$ is within the scope of $\forall$

Note: Anything within a scope of the quantifier can be thought of as a propositional function.

$$\forall x \boxed{\exists y Q(x, y)} \Rightarrow \forall x P(x)$$

$\downarrow$
 $P(x)$

Different combinations of Nested Quantifiers

Order of quantifiers doesn't matter $\left[\begin{array}{l} \forall x \forall y Q(x, y) \\ \forall x \exists y Q(x, y) \\ \exists y \forall x Q(x, y) \\ \exists x \exists y Q(x, y) \end{array} \right] \rightarrow$ Order of quantifiers does matter

Nested Quantifiers (Example 1)

Example 1: Let x and y be the real numbers and $P(x, y)$ denotes " $x + y = 0$."

Find the truth values of

- a) $\forall x \forall y P(x, y)$
- b) $\forall x \exists y P(x, y)$
- c) $\exists y \forall x P(x, y)$
- d) $\exists x \exists y P(x, y)$

Solution: Domain: All real numbers.

a) $\forall x \forall y P(x, y) \equiv \forall x \forall y (x + y = 0)$

"For all real numbers x and y , $x + y = 0$ "

↑ Is it True?

No.

Think of it as "For any combination of x and y , $P(x, y)$."

Example: $1 + 2 \neq 0$

$\forall y \forall x P(x, y) \equiv \forall y \forall x (x + y = 0)$

"For all real numbers y and x , $x + y = 0$ " ← Similar to the previous statement

b) $\forall x \exists y P(x, y) \equiv \forall x \exists y (x + y = 0)$

"For every real number x , there exist a real number y such that $x + y = 0$ "

└ Is it True?

Yes.

You can choose any real number x ; there is always a real number y for that x such that $x + y = 0$.

For example: For

$x = 1$	$x = -1$	$x = \frac{1}{2}$
$y = -1$	$y = 1$	$y = -\frac{1}{2}$

Steps:

1. Consider all different values of x .
2. Find just 1 value of y for each x such that $P(x, y)$ becomes True.

Note: Value of y depends on the value of x .

c) $\exists y \forall x P(x, y) \equiv \exists y \forall x (x + y = 0)$

"There exist some real number y such that for every real number x ,
 $x + y = 0$ "

↑
Is it True?

No.

It is asking us to find a real number y for which $P(x, y)$ becomes true by plugging in every real number x .

What is $P(x, y)$ in our example?

$$P(x, y) \Rightarrow x + y = 0$$

First, take some real number y

Lets take $y = 1$

$$P(x, 1) \Rightarrow x + 1 = 0$$

Now, plug in all real numbers x .

$P(\frac{1}{2}, 1)$ is False, $P(1, 1)$ is False and so on.

No matter what y you choose, $P(x, y)$ is always **False** for all real numbers



What is $P(x, y)$ in our example?

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First, take some real number y

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Now, plug in all real numbers x .

$P(\frac{1}{2}, 1)$ is False, $P(1, 1)$ is False and so on.

No matter what y you choose, $P(x, y)$ is always **False** for all real numbers x .

$$d) \exists x \exists y P(x, y) \equiv \exists x \exists y (x + y = 0)$$

"There exist some real numbers x and y such that $x + y = 0$ "

Obviously. There exist some combination of real numbers x and y exist for which $x + y = 0$.

Take $x = 1$ and $y = -1$ (Many such combinations) $\therefore \exists x \exists y P(x, y)$ is True.

$$1 + (-1) = 0 \text{ is True.}$$

Also, $\exists x \exists y P(x, y) \equiv \exists y \exists x P(x, y)$

Nested Quantifiers (Example 2)

Example 2: Let x and y be the real numbers and $Q(x, y)$ denotes " $x.y = 0$."
Find the truth values of the following:

- a) $\forall x \forall y Q(x, y)$
- b) $\forall x \exists y Q(x, y)$
- c) $\exists y \forall x Q(x, y)$
- d) $\exists x \exists y Q(x, y)$

Solution: a) $\forall x \forall y Q(x, y) \equiv \forall x \forall y (x.y = 0)$

"For all real numbers x and y , $x.y = 0$ "

↑ Is it True? No.

For example: take $x = 1$ and $y = 2$

$$1.2 \neq 0$$

$x.y = 0$ is not satisfied for every combination of real numbers x and y .

Solution: a) $\forall x \forall y Q(x, y) \equiv \forall x \forall y (x.y = 0)$

"For all real numbers x and y , $x.y = 0$ "

└ Is it True? No.

For example: take $x = 1$ and $y = 2$

$$1.2 \neq 0$$

$x.y = 0$ is not satisfied for every combination of real numbers x and y .

b) $\forall x \exists y Q(x, y) \equiv \forall x \exists y (x.y = 0)$

"For every real number x there exist a real number y such that $x.y = 0$ "

└ Is it True? Yes.

Example:

$$\begin{array}{l} x = 1 \\ y = 0 \end{array}$$



$$1.0 = 0$$

$$\begin{array}{l} x = -1 \\ y = 0 \end{array}$$



$$-1.0 = 0$$

$$\begin{array}{l} x = \frac{1}{2} \\ y = 0 \end{array}$$



$$\frac{1}{2}.0 = 0$$

No matter what value of x is chosen, one can always take $y = 0$ to make $x.y = 0$.

c) $\exists y \forall x Q(x, y) \equiv \exists y \forall x (x.y = 0)$

"There exist some real number y such that for every real number x , $x.y = 0$ "

↑ Is it True? **Yes.**

I prefer to convert the above sentence in this form:

Find a real number y such that $Q(x, y)$ becomes true by plugging in all real numbers x .

Step 1: Find a real number y .

Let say $y = 0$.

Step 2: Put y in $Q(x, y)$

$$Q(x, y) = Q(x, 0) \Rightarrow x.0 = 0$$

Step 3: Plug in some values of x in $Q(x, y)$ and get the idea.

It isn't easy to plug in all values of x , but one can get the idea that it is not possible to make $Q(x, y)$ False.

In the above example, we can observe that $x.0$ is always 0 (zero) no matter what value of x could be plugged in.

Step 2: Put y in $Q(x, y)$

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In the above example, we can observe that $x.0$ is always 0 (zero) no matter what value of x could be plugged in.

Therefore, $\exists y \forall x Q(x, y)$ is true.

d) $\exists x \exists y Q(x, y) \equiv \exists x \exists y (x.y = 0)$

"There exist some real numbers x and y such that $x.y = 0$ "

↑ Is it True? Yes.

Take $x = 1, y = 0$.

$$x.y = 0$$

$$1.0 = 0$$

Therefore, $\exists x \exists y Q(x, y)$ is true.

Nested Quantifiers (Example 3)

Example: Let $Q(x, y, z)$ be the statement " $x + y = z$." What are the truth values of the statements $\forall x \forall y \exists z Q(x, y, z)$ and $\exists z \forall x \forall y Q(x, y, z)$, where the domain of all variables consists of all real numbers?

Solution: (i) $\forall x \forall y \exists z Q(x, y, z) \equiv \forall x \forall y \exists z (x + y = z)$

"For all real numbers x and for all real numbers y , there is a real number z such that $x + y = z$ " is true.

$$\begin{aligned}\text{Let } x &= 1 \text{ and } y = -1 \\ z &= 0\end{aligned}$$

It is clear that for every combination of real numbers x and y , there is always a real number z such that $x + y = z$.

$\therefore \forall x \forall y \exists z (x + y = z)$ is True.



(ii) $\exists z \forall x \forall y Q(x, y, z) \equiv \exists z \forall x \forall y (x + y = z)$

"There exist some real number z such that for all real numbers x and y , $x + y = z$ " is true.

Let $z = 2$

Let $z = 2$

For $x = 1$ and $y = 1$, $x + y = z$ is true.

But, for $x = 2$ and $y = 3$, $x + y = z$ is not true.

It is clear that there is no real number z such that $x + y = z$ becomes true for all real numbers x and y .

$\therefore \exists z \forall x \forall y (x + y = z)$ is False.